
1 Counting

There are three types of mathematicians. Those who can count and those who can't.

1.1 Orderings

The details of the question always matter. So always take a second look at what is being asked before leaping.

EXAMPLE 1.1. *Each of twenty students have to present an oral report. In how many orders can this happen?*

The answer is $20 \times 19 \times 18 \dots \times 2 \times 1$.

The most natural way to calculate this is to say: there are 20 choices for who goes first, 19 choices for who goes next, 18 choices for the third person, and so on.

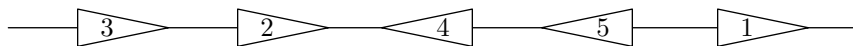
But we can also consider it from the student's perspective. There are 20 slots. There are 20 slots for Adam, then 19 slots for Beth, then 18 slots for Carol, and so on.

The product $20 \times 19 \times 18 \times \dots \times 1$ is called **20 factorial**, written with an exclamation point as $20!$. Note that $1! = 1$. And furthermore, note that $0! = 1$ by definition. (There are good logical reasons why this is the case, mainly that this means that many many formulas still work, but at the end of the day, this is a definition.)

1.2 Sequential Counting

The above is a special case of a general idea in counting: to break up the question into a series of choices. This occurs naturally when counting sequences—a series of objects where the order matters. The idea is that we construct every sequence under consideration by a series of choices. *If we have the same number of options for each choice, then to get the overall number of sequences, multiply the number of options together.*

EXAMPLE 1.2. *A chromosome contains many **genes**. The genes do not overlap, and each gene can be oriented either forward or back. For example, we might depict a chromosome with, in order, gene 3 forward, gene 2 forward, gene 4 back, gene 5 back, gene 1 forward, as follows:*



(a) Given 10 genes, how many possible chromosomes are there, if the 10 genes must appear in order from 1 up to 10?

(b) How many possible chromosomes are there with 10 given genes?

(a) The choices we have are the orientations of the genes. Gene 1 can be oriented forward or back: two choices. Gene 2 can be oriented forward or back: two choices. And so on. So there is a total of 2^{10} possible chromosomes.

(b) The simplest way to do this one, is to first choose the ordering of the genes. There are $10!$ orderings. Then choose the orientations. So the total is $10! \times 2^{10}$.

EXAMPLE 1.3. Define a **word** as any sequence of letters, such as **qjgqri**.

(a) How many 5-letter words are there?

(b) How many 5-letter words end in a consonant (non-vowel)?

(c) How many 5-letter words contain no vowel?

(d) How many 5-letter words have all their letters distinct?

(a) We construct the words one letter at a time. We have 26 options for the first letter. Then, we have 26 options for the second letter. And so. Thus the answer is $26 \times 26 \times 26 \times 26 \times 26 = 26^5$.

(b) Again, we construct the words one letter at a time. We have no constraint on any of the first four letters. For the last letter, there are 5 vowels, and so we have 21 options for the last letter. Thus the answer is $26^4 \times 21$.

(c) This time we have 21 options for each letter. Thus the answer is 21^5 .

(d) We construct the words one letter at a time. As before, there are 26 options for the first letter. For the second letter, there are 25 options (it cannot be the same as the first letter). Now for the key point: when we reach the third letter, there are 2 forbidden letters; no matter what the first two letters are, they are different. And so there are 24 options here. Thus the answer is $26 \times 25 \times 24 \times 23 \times 22$.

► **For you to do!** ◀

1. How many 5-letter words have a middle letter that is a vowel?
2. How many 5-letter words are there whose first and last letter are the same?
3. How many 5-letter words are there whose first and last letter are different?

1.3 The Product Rule and Choosing without Replacement

The fundamental idea for counting discussed above is sometimes summarized by a rule called the **product rule**. The rule is phrased in terms of sets. A **set** is a collection of objects

without repeats. The **size** or **cardinality** of a set S is denoted $|S|$ and is the number of elements in the set.

Lemma 1.1 (The Product Rule) *If A and B are sets, then the set of ordered pairs each consisting of one element of A and one element of B is denoted $A \times B$. This has size:*

$$|A \times B| = |A| \times |B|.$$

There is one ordered pair for each element of A and each element of B .

EXAMPLE 1.4. *Consider a set S with n elements. How many different subsets does S have? (Note that “subset” allows for the possibility of nothing (the empty set) or everything (S itself).)*

The answer is 2^n . We can go through the elements of S one at a time, and for each we have two options: in or out. It is important to note that these choices are independent—one choice does not constrain another.

A common counting situation is what we call **choosing without replacement**: as each item is chosen, it is not resurrected for future choices.

Lemma 1.2 *Consider a universe X of n elements, and $1 \leq k \leq n$.*

- (a) *The number of ways to choose an ordered sequence of k elements from X with replacement is n^k .*
- (b) *The number of ways to choose an ordered sequence of k elements from X without replacement is $n!/(n-k)!$.*

PROOF. (a) Each of the k times we have n choices.

(b) The answer, call it A , is the product $n \times (n-1) \times \cdots \times (n-k+1)$. If we multiply A by $(n-k)!$, we therefore obtain $n!$. That is, $A = n!/(n-k)!$. $\diamond$

Note that, because $0!$ is defined as 1, the second formula works even if $k = n$.

EXAMPLE 1.5. *Consider 4 boys and 5 girls.*

- (a) *In how many ways can they line up for the bus?*
- (b) *In how many ways can they line up for the bus with a girl in front?*
- (c) *In how many ways can they line up for the bus with no two girls next to each other?*

(a) The genders don't matter. There are 9 people. There are 9 options for the person in front, 8 for the next, and so on. The result is $9 \times 8 \times 7 \times \dots \times 1$, which is $9!$.

(b) We have 5 options for the first person. After that there is no constraint. So the answer is $5 \times 8!$.

(c) The only way this can happen is if we alternate girl, boy, girl, boy, girl, et cetera. We have 5 options for the first girl, 4 options for the first boy, and so on. The answer is $5! \times 4!$.

1.4 Counting by Cases or Complements

Another common idea in counting is to divide up the collection into separate cases. In this approach, we count the individual cases separately and then add the answers we get.

EXAMPLE 1.6. *Suppose we roll two dice. If the dice are distinguishable, how many outcomes are there? What if the dice are indistinguishable?*

Say the dice are distinguishable; for example, one red, one green. Then a red 3 and green 5 is considered different to a green 3 and red 5. So there are simply $6^2 = 36$ possibilities, 6 for each die.

Say the dice are indistinguishable. This means that 3 and 5 is considered the same as 5 and 3. We can count these by focusing on the bigger value: there are 6 outcomes where the maximum is 6, 5 outcomes where the maximum is 5, and so on. So the answer is $6 + 5 + 4 + 3 + 2 + 1 = 21$. The pairs are 66, 65, 64, 63, 62, 61, 55, 54, 53, 52, 51, 44, 43, 42, 41, 33, 32, 31, 22, 21, 11

A variation of this is where we count the collection by its **complement**. That is, we count the collection by counting the whole universe and then subtracting those things that are **not** in our collection.

EXAMPLE 1.7. *How many 5-letter words have at least one vowel?*

Well, we could try to count those with one vowel, those with two vowels, and so on. But it is easier to count those with no vowel and subtract that from the total. We calculated these two quantities in Example 1.3. So the answer is $26^5 - 21^5$.

Some people like to summarize this approach with the sum rule:

Lemma 1.3 (The Sum Rule) *If sets A and B are disjoint (meaning their intersection is empty), then $|A \cup B| = |A| + |B|$. More generally,*

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

PROOF. If we count the elements in A and B , then we get every element in the union $A \cup B$, except that the elements in the intersection $A \cap B$ are counted twice. $\diamond$

We use the product and sum rules all the time in counting, even if we often don't make this explicit.

► **For you to do!** ◀

4. In the Mighty Math Mob there are 4 freshman, 5 sophomores, 6 juniors, and 7 seniors. In how many ways can one form a committee of 3 people, no two of whom are in the same year?

Exercises

- 1.1. At the local BananaAnts restaurant, there is a special three-course dinner. You choose one appetizer, one entree, and one dessert; there are 3 appetizers, 4 entrees and 5 desserts. How many possible dinners are there?
- 1.2. Consider South Carolina's license plate, which is three letters followed by a three-digit number. (Leave answers as products.)
 - (a) How many possible license plates are there if the number must be even?
 - (b) How many possible license plates are there if no vowels allowed?
 - (c) How many possible license plates are there if all letters and digits must be distinct?
 - (d) How many possible license plates are there if we change the rules to allow either 3 or 4 letters?
- 1.3. An SSN is a 9-digit number with zeroes allowed in every position.
 - (a) How many SSNs are there all of whose digits are even?
 - (b) How many SSNs are a multiple of 2?
 - (c) How many SSNs are there that are palindromes (read the same forward as backwards)?
 - (d) How many SSNs are there all of whose digits are distinct?
 - (e) How many SSNs are there such that no two consecutive digits are the same?
 - (f) How many SSNs are there whose digits are strictly increasing.
- 1.4. Wendy needs to schedule 10 different speakers at a rally (who each speak exactly once).
 - (a) How many orderings are there such that Fred speaks after Beth (not necessarily consecutive)?

- (b) How many orderings are there such that Fred speaks immediately after Beth?
 - (c) There are 3 Democrats and 7 Republicans on the list. How many orderings are there with a Democrat as an opening speaker and a Republican as a closing speaker?
- 1.5. There are six different Welsh books, eight different Xhosa books, and five different Yiddish books.
- (a) In how many different ways can these books be arranged on a bookshelf?
 - (b) In how many ways can these books be arranged on a bookshelf if all books in the same language are grouped together?
- 1.6. Consider 5-letter words again.
- (a) How many start and finish with a vowel?
 - (b) How many have at most one vowel?
 - (c) How many have exactly three vowels with no two vowels next to each other?
- 1.7. Consider 5 boys and 5 girls.
- (a) In how many ways can they line up for the bus with a girl in front?
 - (b) In how many ways can they line up for the bus with both a girl in front and at the back?
 - (c) In how many ways can they line up for the bus with a girl in front or at the back or both?
 - (d) In how many ways can they line up for the bus with no two girls next to each other?
- 1.8. A **pandigital** number is a 9-digit number all of whose digits are different and containing no zero. (For some of these, it might be quicker to write some code.)
- (a) How many pandigital numbers are there?
 - (b) How many pandigital numbers are even?
 - (c) How many pandigital numbers are perfect squares?
 - (d) How many pandigital numbers are prime numbers?
- 1.9. Prove the following generalization of the Sum Rule: For all sets A, B, C ,

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

1.10. If we write out $30!$, how many zeroes are there at the end?

Solutions to Practice Exercises

1. There are 5 choices for the middle letter and 26 choices for each of the other letters. So answer is 5×26^4 .
2. Once we choose the first letter, the last letter is determined. So answer is 26^4 .
3. $26^4 \times 25$.
4. First choose which year is not represented, then choose the representatives. So answer is $4 \times 5 \times 6 + 4 \times 5 \times 7 + 4 \times 6 \times 7 + 5 \times 6 \times 7 = 638$.